

# Deepak Mallampati

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US-based (Kent, OH) - F-1 OPT, US-work-authorized; available for SF relocation

## PROFESSIONAL SUMMARY

Full-stack Applied AI engineer shipping production AI inside regulated enterprise systems. At Energy Solutions: hardened T-SQL stored procedures for an oil & gas Order-to-Cash platform (contracts, nominations, allocations, invoicing), built Jenkins CI/CD with rollback checkpoints (>30% fewer deployment errors), and supported RBAC + audit logging in a compliance-sensitive environment. At Progress Solutions: ships RAG + agentic LLM services as FastAPI/Docker stacks with structured logging (~30% post-deploy defect reduction). Full-stack TypeScript + React experience from Dream Decoder (FastAPI + React/TS/Vite/Tailwind) and Manga Lens (MV3 Chrome extension, shipped). Available for SF on-site; F-1 OPT, US-work-authorized today.

## CORE COMPETENCIES

Forward Deployed Engineering • Full-Stack TypeScript / React / Next.js-adjacent • PostgreSQL / SQL / T-SQL • Enterprise Integrations (ERP, RBAC, Audit) • Docker / Jenkins / CI-CD • LLM Application Development • Customer-Embedded Delivery • 0 → 1 Founding Builder

## PROFESSIONAL EXPERIENCE

**Jr. AI/ML Engineer Trainee** - Progress Solutions Inc. *Jul 2025 - Present* | USA | Healthcare Technology, AI Consulting, SaaS  
Built **Retrieval-Augmented Generation (RAG) systems** for clinical knowledge retrieval and healthcare documentation search; recursive semantic chunking + transformer-based embeddings; improved retrieval precision by ~35%, reduced irrelevant retrieval by >30%, grounded response alignment exceeded 90%.

Developed **agentic LLM workflows** for multi-step healthcare queries with structured reasoning, tool discipline, and grounding rules; improved agent response stability by ~25%.

Shipped **predictive ML pipelines** (scikit-learn, XGBoost) for patient no-show prediction, care engagement scoring; improved recall by 15-20% on high-risk categories while holding precision >90% via class weighting and threshold calibration.

Packaged ML/LLM inference as **FastAPI/Flask** RESTful services, containerized with **Docker**, with structured logging and load simulation; reduced post-deployment defects by ~30%.

Built preprocessing pipelines (Pandas, NumPy) for EHR extracts, appointment histories, support ticket logs; raised dataset reliability above 98% and cut downstream model instability by ~40%.

Maintained HIPAA-conscious data governance: de-identification, data lineage, evaluation audit trails.

**Database & DevOps Performance Engineer (Intern)** - Energy Solutions International (Emerson) *Jun 2022 - Apr 2023* | Hyderabad, India | Oil & Gas, Enterprise ERP, Industrial Automation

Optimized SQL and **T-SQL stored procedures** powering the Synthesis Order-to-Cash platform (contracts, nominations, allocations, invoicing); reduced query time ~20% and latency ~25%.

Built **CI/CD pipelines with Jenkins** for schema updates and stored-procedure deployments; reduced deployment errors >30%, cut release cycle time ~35-40%.

Designed performance dashboards (SQL Server DMVs + Grafana) to track long-running queries, deadlocks, CPU/I/O contention; cut incident recurrence ~25%, improved RCA ~30%.

Tuned database maintenance (index rebuilds, statistics updates, partition-aware indexing) and reconciliation queries; improved reconciliation runtime ~15%, reduced deadlocks ~15%.

Supported RBAC implementation and audit logging for financial modules in a compliance-sensitive oil & gas environment.

**Machine Learning Engineer** - Suvidha Foundation *Jan 2022 - Mar 2022* | Hyderabad, India | Nonprofit Tech, Video Analytics, Applied NLP

Built a **transformer-based video summarization and highlight-generation system** across 5,000+ sessions; transcript preprocessing, hierarchical summarization for long-context, and timestamp-aligned clip extraction reduced manual review time by 60-70%.

Achieved ~85% **alignment** between AI-selected highlights and human-curated key moments.

Implemented **document Q&A with RAG** (semantic chunking + embedding retrieval); reduced hallucinated outputs ~30% and raised contextual answer accuracy >85%.

Deployed the stack as a **Flask-based API** with a lightweight web interface for non-technical staff.

## SELECTED PROJECTS

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**Manga Lens** - Chrome Extension for Real-Time AI Manga/Webtoon Translation (*shipped, Chrome Web Store*)

Full-stack browser extension - Manifest V3, content scripts, service workers, `captureVisibleTab` for panel capture. Integrates **four AI vision providers** (Claude Sonnet, GPT-4o mini, GPT-4.1 Nano, Ollama/minicpm-v local). Multi-section capture pipeline handles tall webtoon panels via viewport-slice screenshots with coordinate remapping and dedup. Seven-day translation cache, per-domain selector configs for 29 sites, narrowed host permissions.

**Agentic Healthcare Claims Processing & Fraud Risk Intelligence System**

Multi-agent pipeline (Intake Normalization → Validation → Consistency Analysis → Duplicate Detection → Fraud Risk Scoring) with **schema-validated JSON contracts between agents** to prevent cascading hallucinations. RAG-grounded CPT/ICD validation against vector-indexed policy documents; approximate-nearest-neighbor similarity search for duplicate claim detection; explainable risk scoring with audit-ready reasoning traces.

**Driver Drowsiness Detection System** - Real-Time YOLOv8 Fatigue Monitoring

Replaced a two-stage CNN pipeline with a **unified YOLOv8** detection-and-classification model; reduced inference latency ~**30%**. Augmentation for lighting/head-pose robustness, sliding-window confidence aggregation to suppress blink-driven false positives (~**25% reduction**), adaptive frame skipping, and NMS tuning for stable real-time operation via OpenCV.

**Agentic Pixel Character Synthesis & Animation Engine** (*ongoing*)

Phased generative AI system for identity-consistent, pose-controlled pixel character synthesis with sprite-sheet export. Stable Diffusion fine-tuning + LoRA, ControlNet + OpenPose/MediaPipe for pose conditioning, and an **LLM-based agent orchestrator**. Backend: PyTorch, Hugging Face Diffusers, FastAPI, Docker.

**Dream Decoder** - Multimodal Creative Intelligence Platform

Full-stack FastAPI + React/TypeScript/Vite/Tailwind app coordinating multimodal APIs (Perplexity Sonar for interpretation, Replicate image models). Introduced **intermediate structured prompt transformation layers** - improved contextual alignment ~**30%** and first-pass image success rate ~**25-30%**.

## EDUCATION

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**Master's Degree** - Kent State University, Ohio, USA *Aug 2023 - May 2025*

## WORK AUTHORIZATION

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F-1 OPT - work-authorized in the United States; future sponsorship required.

## TECHNICAL SKILLS

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**Languages & Frameworks:** Python, FastAPI, Flask, SQL (T-SQL, PostgreSQL, SQLite), TypeScript, React, C++ (Arduino), PHP

**LLM & GenAI:** LLM application development, RAG, agentic workflows, structured outputs, prompt engineering, evaluation pipelines, guardrails, grounding, embeddings, vector search, responsible AI

**ML Libraries:** scikit-learn, XGBoost, PyTorch, Hugging Face Transformers, Diffusers, LangChain, LlamaIndex

**Computer Vision & Multimodal:** YOLOv8, OpenCV, ControlNet, OpenPose/MediaPipe, Stable Diffusion, video analytics

**Data & Analytics:** Pandas, NumPy, feature engineering, time-series forecasting, walk-forward validation, NetworkX

**Infra & DevOps:** Docker, Jenkins, Grafana, CI/CD, MLOps/LLMOps, RESTful APIs, observability/logging

**Domains:** Healthcare AI (HIPAA-conscious), Applied AI, Enterprise AI, Workflow Automation, Computer Vision, Multimodal Systems